Model
designer & workload

Design

Software stack + Silicon

Train & serve

Successor model
runs on what it built

Recurse

Self-scaling intelligence

the successor becomes the next designer

Co-design → Co-evolution (▷ §2–§3)

- **Problem**: Fixed target → Moving target
- **Executor**: People → Model
- **Feedback**: One-shot → S1→S6 ladder
- **Autonomy**: L0 assist → L4 recursive

And more...

Model → Silicon

- RTL generation
- Verification & debug
- EDA tools, end-to-end

Model → Stack

- LLM-written kernels
- Compilers
- Serving & deployment

Silicon → Model

- Attention & memory
- Sparsity & MoE
- Numeric formats

spec → RTL → GDSII · kernel → profile

AI-driven · L0–L2 · reaches L3

vs

HBM · FP8 · 2:4 · MoE, carried up by people

Design Flows Down, Constraints Flow Up (▷ §4)

Industrial Precedents

- RL adders, layouts
- Chip placement
- Algorithm discovery

Recursive Loop (L4)

- Circuit → next TPU
- Kernel → training
- Human sign-off

Human-led Co-design

- Workload-driven ASIC
- Model built for NPU
- Claims, undisclosed

AI-designed circuits in AI hardware

PrefixRL · AlphaChip · AlphaDev

designer ≠ trained model
signal not connected · gain too small

Learn

AlphaEvolve

TPU

trains the next Gemini

Closed Loops and Where They Break (▷ §5)

Pre-training

- Domain-adaptive data
- Synthetic, anchored
- Tacit, cross-layer

Post-training

- Verifiable rewards
- Formal checks
- Reward hacking

Agents

- Horizon & memory
- Multi-fidelity S1–S6
- Cross-layer attribution

knowledge → Support → reward → Support → agent → Needs → world model

optimizers find verifier gaps

silicon as calibration

Intelligence for the Loop (▷ §6)

Exam benches (one layer)

RTL	Verification	Physical	Kernels	Living
VerilogEval · CVDP	FVEval	ChiPBench	KernelBench	NotSoTiny

Training grounds (toward L3)

Environments	AI R&D	HW/SW	Cross-layer	Loop period
SWE-Gym · MLGym	RE-Bench	HSCO-Bench	joint > sum	time per turn

Benchmarks: From Exam Hall to Training Ground (▷ §7)

Open Problems (▷ §8)

- Physics: proxy worlds, faster chips
- Intelligence: world models, memory
- Ecosystem: open data, shared gyms
- Beyond: self-expansion (L5–L6)